

Usma Niyaz

Email: usma.20csz0015@iitrpr.ac.in, niyazus@gmail.com

GitHub: <https://github.com/UsmaBhat>

Website: <https://usmabhat.github.io/UsmaNiyazBhat/>

INTERESTS

Medical Image Analysis, Model Compression, Knowledge Distillation, Deep Learning, Machine Learning, and Computer Vision

EDUCATION

IIT Ropar PhD Computer Science & Engineering GPA: <i>8.07</i>	Punjab, India currently
Central University of Jammu Master of Technology (M.Tech.) Computer Science & Technology GPA: <i>8.92</i>	J&K, India 2016 - 2018
University of Kashmir Bachelor of Technology (B.Tech.) Computer Science & Engineering <i>81.6%</i>	J&K, India 2011 - 2015

TEACHING EXPERIENCE

Assistant Professor(Contractual), Department of Computer Science and Engineering, NIT Srinagar, India	Sept 2018- Dec 2018
Assistant Professor(Contractual), Department of Information and Technology, NIT Srinagar, India	March 2019-2020

PH.D COURSES

	GRADE POINTS
Artificial Intelligence	9
Machine Learning	8
Artificial Neural Networks	8
Advanced Computer Vision	9
Digital Image Processing and Analysis	7

TEACHING ASSISTANT

Artificial Neural Networks (CS504), Optimization for Machine Learning (CS560), Programming Paradigms & Pragmatics (CS202), Computer Graphics (CS515), Digital Image Processing & Analysis (CS517), Data Structures and Algorithms (CS506)

ADD-ON COURSES

Computer Vision	Stanford (2018)
Machine Learning - Stanford	Coursera (2017)

COLLABORATION

Mom2b: Predicting peripartum depression using a smartphone application and digital phenotyping, Uppsala University, Sweden

Omics data in parinatal mental health, Uppsala University, Sweden

ANRF-CRG project entitled “Characterization of Perfusion Deficit in Stroke using Resting-State Functional Magnetic Resonance Imaging (fMRI)”

JOURNAL PUBLICATIONS

SHAPER: Shape-aware Parameter-Efficient Representation Learning for Medical Image Segmentation

Usma Niyaz, Abhishek Singh Sambyal, Ranjana Roy Chowdhury, Deepti R. Bathula
Knowledge Based Systems (SCI; IF: 7.6), 2026

Prototypical Aggregate Network - Boosting Few-Shot Learning for Medical Image Classification

Ranjana Roy Chowdhury, Usma Niyaz, Deepti R. Bathula
Multimedia Tools and Applications (SCI; Q1), 2026.

MRI-to-PET Cross-Modality Translation using Globally & Locally Aware GAN (GLA-GAN) for Multi-Modal Diagnosis of Alzheimer’s Disease

Apoorva Sikka, Skand Peri, Jitender Singh Virk, Usma Niyaz, Deepti R. Bathula

The Journal of Precision Medicine: Health and Disease, 2025

Leveraging Different Learning Styles for Improved Knowledge Distillation in Biomedical Imaging

Usma Niyaz, *Abhishek Singh Sambyal, Deepti R. Bathula*

Computers in Biology and Medicine (SCI; IF: 7.7), 2024

Understanding Calibration of Deep Neural Networks for Medical Image Classification

Abhishek Singh Sambyal, Usma Niyaz, Narayanan C. Krishnan, Deepti R. Bathula

Computer Methods and Programs in Biomedicine (SCI; IF: 6.1), 2023

CONFERENCE PUBLICATIONS

ShapeDistill: Shape-Constrained Knowledge Distillation for Medical Segmentation

Usma Niyaz, *Deepti R. Bathula*

18th International Conference on Machine Vision (ICMV), 2025.

Dual-Level Adaptive Sampling for Enhanced Few-Shot Medical Image Classification

Ranjana Roy Chowdhury, Usma Niyaz, Deepti R. Bathula

7th IEEE Symposium on Computers & Informatics (ISCI), 2025)

LS+: Informed Label Smoothing for Improving Calibration in Medical Image Classification

Abhishek Singh Sambyal, Usma Niyaz, Deepti R. Bathula

27th International Conference on Medical Image Computing and Computer Assisted Intervention, (MICCAI), 2024

Wavelet-Based Feature Compression for Improved Knowledge Distillation

Usma Niyaz, *Abhishek Singh Sambyal, Deepti R. Bathula*

IEEE 21st International Symposium on Biomedical Imaging (ISBI), 2024

Class-Wise Feature Map Selection Based Prototypical Networks

Ranjana Roy Choudhary, Usma Niyaz, Deepti R. Bathula

IEEE 21st International Symposium on Biomedical Imaging (ISBI) Satellite Workshop, 2024

Augmenting Knowledge Distillation with Peer-to-Peer Mutual Learning for Model Compression

Usma Niyaz, *Deepti R. Bathula*

IEEE 19th International Symposium on Biomedical Imaging (ISBI), 2022

Evaluation of Deep Learning model with Optimizing and Satisficing metrics for Lung Segmentation

Usma Niyaz, *Abhishek Singh Sambyal, Devanand*

International Conference on Soft Computing for Problem Solving (SocProS), 2018

Advances in Deep Learning Techniques for Medical Image Analysis

Usma Niyaz, *Abhishek Singh Sambyal, Devanand*

IEEE International Conference on Parallel, Distributed and Grid Computing (PDGC), 2018

REVIEWER

Medical Image Computing and Computer Assisted Intervention (MICCAI), International Symposium on Biomedical Imaging (ISBI) Computers in Biology and Medicine, Knowledge Based Systems, Computers and Electrical Engineering, Computer Vision & Image Processing (CVIP)

CONFERENCES/SYMPOSIUM/WORKSHOPS

18th International Conference on Machine Vision 2025

Paris, France

19-22 October 2025

8th International Conference on Data Science and Management of Data (CODS-COMAD)

IIT Jodhpur, India

18-21 December 2024

27th International Conference on Medical Image Computing and Computer Assisted Intervention, (MICCAI) 2024

Marrakesh, Morocco
6-10 October 2024

21st IEEE International Symposium on Biomedical Imaging (ISBI) 2024
Athens, Greece
27-30 May 2024

The Indian Symposium on Machine Learning (IndoML) 2023
IIT Bombay, Maharashtra, India
21-23 December 2023

19th IEEE International Symposium on Biomedical Imaging (ISBI) 2022
ITC Royal Bengal, Kolkata, India
28-31 March 2022

24th International Conference on Medical Image Computing and Computer Assisted Intervention, (MICCAI) 2021.
27th September - 1st October 2021

Google Research India-Graduate Symposium 2021
7-10 April 2021

8th International Conference on Soft Computing for Problem Solving (SocProS) 2018
Vellore Institute of Technology Vellore, India
17-19 December, 2018

5th International Conference on Parallel, Distributed and Grid Computing (PDGC) 2018
Solan Himachal Pradesh, India
20-22 December, 2018

International Conference on Recent Innovations in Computing (ICRIC 2018)
Central University of Jammu
05 - 06 March, 2018

Workshop: Recent Trends in Computer Technology 2018
Central University of Jammu
21 - 28 October, 2018

PROJECTS

Lung Segmentation [Python]

In the Kaggle competition "*Finding and Measuring Lungs in CT Data*," we proposed a novel CNN-based architecture called FastSegmentation (FS). Unlike traditional architectures such as U-Net, FS eliminates skip connections, focusing on efficient evaluation of deep learning models with optimizing and satisfying performance metrics for lung segmentation.

Lung Cancer Detection using CNN [Python]

For my M.Tech. project, I designed and implemented a method for lung cancer classification by analyzing thousands of CT scan images. This approach utilized a Convolutional Neural Network (CNN) architecture, which was developed from scratch, to achieve accurate and reliable results..

Copy At Once [C#]

Developed a Windows application designed to copy files and folders simultaneously to all connected removable devices. The application aimed to save time by overcoming the default Windows limitation, which allows file transfers to only one drive at a time.

Traffic Analysis System for Smart Cities [Arduino]

The project focused on implementing a method for detecting and analyzing traffic jams on roads by establishing a network of interconnected vehicles. This approach enabled real-time traffic monitoring and analysis to improve road efficiency and reduce congestion.

PROJECTS GUIDED

Twitter bot detection [Python]

Developed a machine learning-based approach for detecting bot accounts using the Naive Bayes algorithm. The project focused on analyzing account behaviors and patterns to accurately distinguish between genuine users and bots.

Violence Detection Using Deep Learning [Python]

Designed and developed a technique for the automatic analysis of surveillance videos to detect the

presence of violence. The approach leveraged 3D Convolutional Neural Networks (3D CNNs) to capture spatial and temporal features, enabling accurate identification of violent activities in real-time video streams.

Vehicle Simulation and Object detection in Gaming Environment(Phase I) [Python]

Developed a simulation for a vehicle that simultaneously detects lanes and objects on the road. The system integrates advanced computer vision techniques to identify lane markings and objects in real-time, enabling enhanced navigation and safety features for autonomous vehicles. The project is still in Phase 1

ACHIEVEMENTS

1. **Society of Women Engineers (SWE)** Received scholarship for the 2025–2026 academic year.
2. **Anusandhan National Research Foundation (ANRF) Travel Grant (formerly SERB)**
Awarded the travel grant to present my paper at the 21st IEEE International Symposium on Biomedical Imaging (ISBI) 2024.
3. **ISBI Travel Grant** Received the travel grant to attend 21st IEEE International Symposium on Biomedical Imaging (ISBI) 2024.
4. **Lead, "Pehachaan Ek Safar" NGO at IIT Ropar:** Served as a Lead in the "Pehachaan Ek Safar" NGO at IIT Ropar, Punjab, India, where we teach underprivileged children and assisted them in securing admissions to government and private schools.
5. **Gate Qualified:** Successfully qualified the Graduate Aptitude Test in Engineering (GATE) 2020.
6. **Best Paper Presentation Award, ICRIC 2018:**
7. **Best Paper Award, SOCPROS 2018:**
8. **National Mobile Championship - RoboMania2013:** Qualified the prelims round in the National Mobile Championship "RoboMania2013" held at the Islamic University of Science and Technology, organized by Robonext.
9. **Skill Development Programme on CCNA, University of Kashmir:** Successfully completed a 2-month Skill Development Programme on CCNA organized by the Entrepreneurship Development Cell (EDC) at the University of Kashmir.

MAILING ADDRESS

Usma Niyaz Bhat,
H-No.7 Lane No.9, Momin Abad,
Srinagar, J&K, 190009
India